Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 11×3

2. 11×7

3. 11×4

4. 11×9

5. Reflect (-3, 4) across the y-axis.

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. 5×2

8. Find the mean of 1, 4, 5, 6.

9. Find the mean of 2, 3, 7, 8.

10. Find the median of 9, 1, 5, 3, 7.

11. Find the median of 8, 2, 4, 6.

12. Four scores have mean 5. Three are 2, 4, 7. Find the fourth.

13. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

14. Choose the statistical question: (A) How many pages are in one named book? (B) How many pages are in each of 10 books? Explain.

15. Make a dot plot for 3, 3, 4, 5, 5, 5, 7. State the total dots and most frequent value.

16. Find histogram frequencies for 2, 3, 4, 4, 5, 6, 7, 7 using bins 2 to under 4, 4 to under 6, and 6 to under 8.

17. Find the mean of 4, 6, 8, 10. Explain it as a fair share.

18. Order 10, 4, 7, 6, 7, 11 and find the median.

CHALLENGE / Explain your plan and check the result

C1. Two teams record scores. Team A has three scores with mean 8; Team B has five scores with mean 12. Find the combined mean. Explain why simply averaging 8 and 12 gives the wrong result.

C2. Create four nonnegative whole-number data values with mean 5 and median 5, but range 8. Give one set and verify all three conditions. Then give a different set with the same mean and median but smaller range.